Year 10 Mathematics

Shape and Space 1

5.5 Circles

Circles

STARTER · 5 MIN

Date: _____ **Title:** Circles

LEARNING OBJECTIVES

- Find the circumference of a circle
- Find the area of a circle
- Solve problems with semicircles and give answers in terms of π

STARTER *Answer in your book. No calculators.*

1. Work out 2×7
2. Work out 5^2
3. Work out 3.14×10
4. Work out 3.14×25

Starter — Answers

STARTER

Tick or correct in green pen

1. $2 \times 7 = \mathbf{14}$
2. $5^2 = \mathbf{25}$
3. $3.14 \times 10 = \mathbf{31.4}$
4. $3.14 \times 25 = \mathbf{78.5}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the circumference of a circle	☐
SILVER	I can find the area of a circle	☐
GOLD	I can solve problems with semicircles and give answers in terms of π	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Circles

INTRODUCTION · 3 MIN

The diameter is twice the radius: $d = 2r$

Circumference

The distance around the circle.

$C = \pi d$ or $C = 2\pi r$

Area

The space inside the circle.

$A = \pi r^2$

Use the π button

Give answers to 3 significant figures.

$\pi \times 8 = 25.1$ (3 s.f.)

In terms of π

Leave π in the answer.

$r = 3$: $A = 9\pi$

Examples 1 and 2 — Circumference and Area

EXAMPLES · 5 MIN

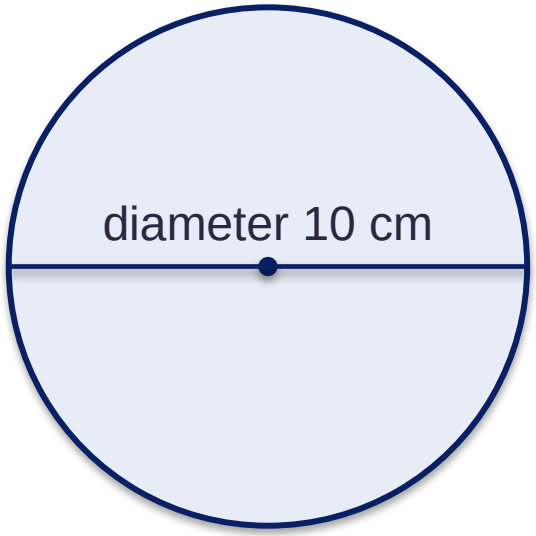
BRONZE

Watch and copy into your book

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

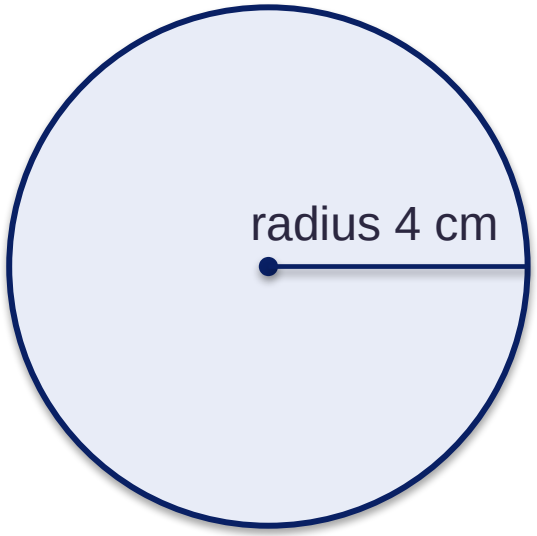
Example 1

Find the circumference. Give your answer to 3 s.f.



Example 2

Find the area. Give your answer to 3 s.f.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$C = \pi \times 10$

$C = 31.4$ cm

Tip: $C = \pi d$

Example 2

$A = \pi \times 4^2$

$A = 50.3$ cm²

Tip: square the radius first.

Task 1

PRACTICE · 9 MIN

BRONZE

Give your answers to 3 significant figures

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

1. Circumference: diameter 8 cm

2. Circumference: radius 5 cm

3. Area: radius 6 cm

4. Area: diameter 10 cm

5. Circumference: radius 3.5 cm

★ Area of a circle with radius 3 cm, in terms of π

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $C = \pi \times 8 = \mathbf{25.1\ cm}$

2 $C = 2 \times \pi \times 5 = \mathbf{31.4\ cm}$

3 $A = \pi \times 6^2 = \mathbf{113\ cm^2}$

4 $A = \pi \times 5^2 = \mathbf{78.5\ cm^2}$

5 $C = 2 \times \pi \times 3.5 = \mathbf{22.0\ cm}$

★ $A = \pi \times 3^2 = \mathbf{9\pi\ cm^2}$

Examples 3 and 4 — Semicircles and π

EXAMPLES · 5 MIN

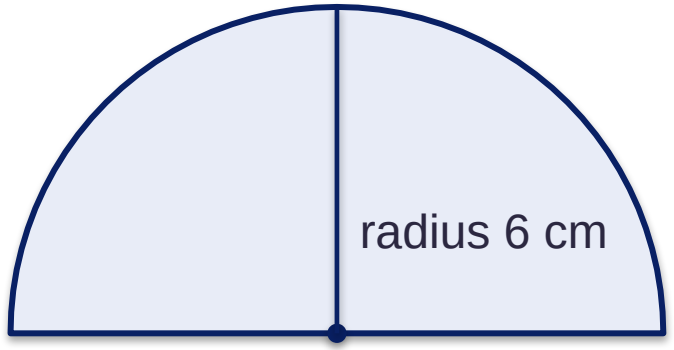
SILVER

Watch and copy into your book

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

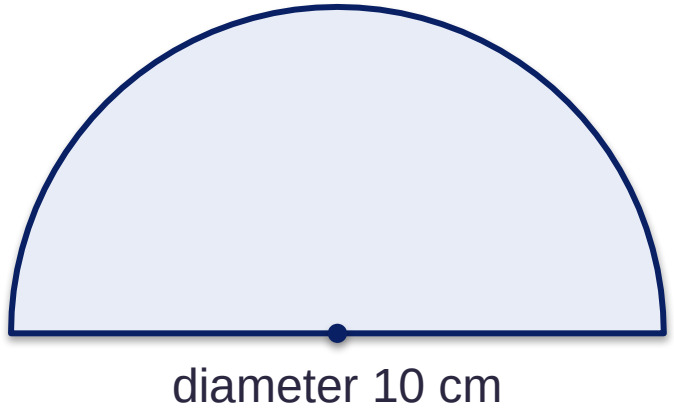
Example 3

Find the area of the semicircle. Give your answer in terms of π.



Example 4

Find the perimeter of the semicircle. Give your answer to 3 s.f.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$A = \frac{1}{2} \times \pi \times 6^2$$

$A = 18\pi$ cm²

Tip: a semicircle is half a circle.

Example 4

$$\text{Arc} = \frac{1}{2} \times \pi \times 10$$

$$= 15.7$$

$$P = 15.7 + 10$$

$P = 25.7$ cm

Tip: include the straight edge.

Task 2

PRACTICE · 9 MIN

SILVER

Give answers to 3 s.f. unless asked for π

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

1. Area of a semicircle with radius 4 cm

2. Perimeter of a semicircle with diameter 12 cm

3. Area of a quarter circle with radius 10 cm

4. Circumference of a circle with diameter 9 cm, in terms of π

5. Area of a circle with radius 7 cm, in terms of π

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $A = \frac{1}{2} \times \pi \times 4^2 = \mathbf{25.1\ cm^2}$

2. $P = \frac{1}{2} \times \pi \times 12 + 12 = \mathbf{30.8\ cm}$

3. $A = \frac{1}{4} \times \pi \times 10^2 = \mathbf{78.5\ cm^2}$

4. $C = \pi \times 9 = \mathbf{9\pi\ cm}$

5. $A = \pi \times 7^2 = \mathbf{49\pi\ cm^2}$

Examples 5 and 6 — Working Backwards

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

Example 5

(2 marks)

A circle has a circumference of 50 cm.

Find its radius.

Give your answer to 3 s.f.

Example 6

(3 marks)

A circle has an area of 100 cm^2.

Find its radius.

Give your answer to 3 s.f.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$2\pi r = 50$$

$$r = \frac{50}{2\pi}$$

r = 7.96 cm

Tip: divide by 2π.

Example 6

$$\pi r^2 = 100$$

$$r^2 = \frac{100}{\pi} = 31.83$$

r = 5.64 cm

Tip: divide by π, then square root.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

1. A circle has circumference 30 cm. Find its diameter. Give your answer to 3 s.f. (2 marks)

2. A circle has area 80 cm^2. Find its radius to 3 s.f. (3 marks)

3. A wheel has diameter 60 cm. How far does it travel in 100 turns? Give your answer in metres to 3 s.f. (3 marks)

4. A circle has area 64π cm^2. Find its circumference in terms of π. (3 marks)

5. A circle fits exactly inside a square of side 10 cm. Find the area outside the circle to 3 s.f. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $d = 30 \div \pi = \mathbf{9.55\ cm}$

2. $r^2 = 80 \div \pi = 25.46, \quad r = \mathbf{5.05\ cm}$

3. $100 \times \pi \times 60 = 18\,850\ cm = \mathbf{188\ m}$

4. $r^2 = 64, \quad r = 8, \quad C = \mathbf{16\pi\ cm}$

5. $100 - \pi \times 5^2 = 100 - 78.5 = \mathbf{21.5\ cm^2}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the circumference of a circle with diameter 6 cm.

SILVER Find the area of a circle with radius 9 cm.

GOLD A circle has area 50 cm^2. Find its radius.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $C = \pi \times 6 = \mathbf{18.8\ cm}$

SILVER $A = \pi \times 9^2 = \mathbf{254\ cm^2}$

GOLD $r^2 = 50 \div \pi = 15.92, \quad r = \mathbf{3.99\ cm}$

Exam Practice — Circumference and Area

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

Question 1

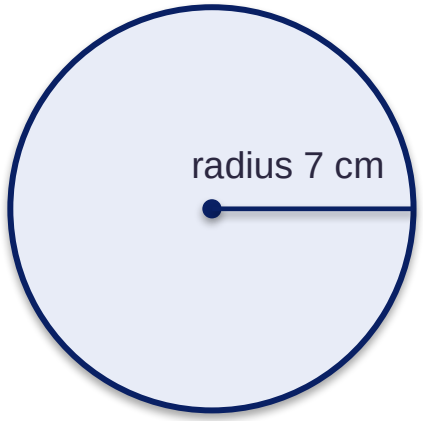
(2 marks)

The radius of the circle is 7 cm.

Work out the circumference.

Give your answer to 3 s.f.

Diagram NOT accurately drawn



..... cm

Question 2

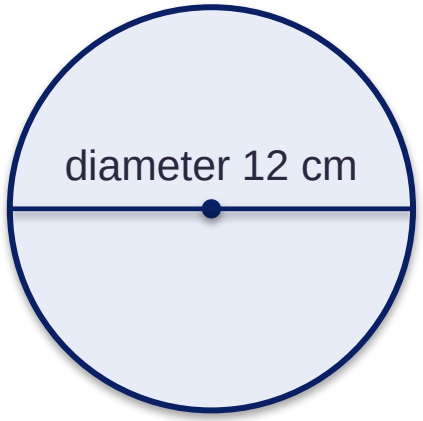
(2 marks)

The diameter of the circle is 12 cm.

Work out the area of the circle.

Give your answer to 3 s.f.

Diagram NOT accurately drawn



..... cm^2

Exam Practice — Semicircles and π

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

Question 3

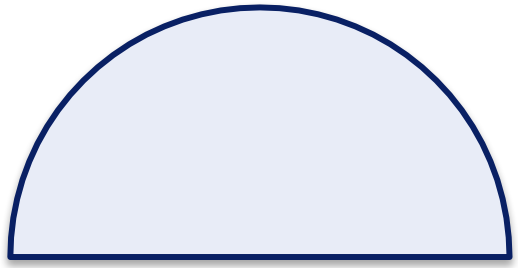
(3 marks)

A semicircle has diameter 10 cm.

Work out its perimeter.

Give your answer to 3 s.f.

Diagram NOT accurately drawn



diameter 10 cm

..... cm

Question 4

(2 marks)

A circle has radius 6 cm.

Give each answer in terms of π.

(a) Work out the area. (1)

..... cm^2

(b) Work out the circumference. (1)

..... cm

Exam Practice — Working Backwards

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $C = \pi d = 2\pi r$ $A = \pi r^2$

Question 5

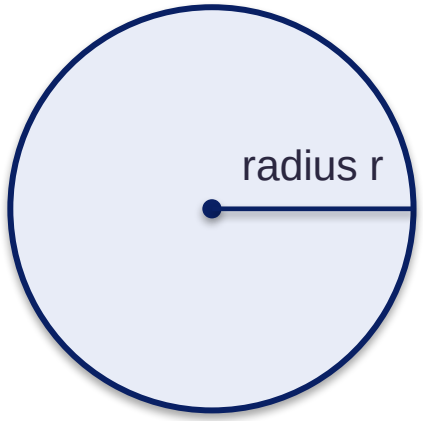
(3 marks)

A circle has area 200 cm^2.

Work out the radius of the circle.

Give your answer to 3 s.f.

Diagram NOT accurately drawn



..... cm

Question 6

(4 marks)

A circle has circumference 50 cm.

Work out the area of the circle.

Give your answer to 3 s.f.

..... cm^2

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$2 \times \pi \times 7$ — M1

$= 43.98\ldots$

44.0 cm — A1

Tip: C = 2πr or πd.

Question 2

$r = 6$

$\pi \times 6^2$ — M1

$= 113\ \text{cm}^2$ — A1

Tip: halve the diameter before squaring.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

π × 10 ÷ 2 = 15.70…	M1
15.70… + 10	M1
= 25.7 cm	A1

Tip: perimeter includes the straight edge.

Question 4

(a) 36π cm²	B1
(b) 12π cm	B1

Tip: 'in terms of π' means leave π in.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$\pi r^2 = 200$	M1
$r^2 = 200 \div \pi = 63.66\ldots$	M1
$r = 7.98$ cm	A1

Tip: undo × π first, then square root.

Question 6

$2\pi r = 50$	M1
$r = 7.957\ldots$	A1
$\pi \times 7.957\ldots^2$	M1
$= 199$ cm^2	A1

Tip: keep the unrounded r in your calculator.